# Exponentials and Logarithms

Questions 1–55

**Question 1.** Solve the following equations, writing your answer in logarithmic form:

a) $4^x = 3$
b) $3^x = 4$
c) $2^x = \frac{1}{3}$
d) $9^x = \frac{1}{100}$

**Question 2.** Solve the following equations, writing your answer in logarithmic form:

a) $2^{x+1} = 3$
b) $3^{x-5} = \frac{1}{5}$
c) $5^{2x-1} = 7^{500}$
d) $6^{3x+4} = 9^{250}$

**Question 3.** Solve the equation

$$6^{x^2-5} = \frac{1}{216}$$

giving your answers in exact form.

**Question 4.** Solve the equation

$$8^{10-x^2} = 32768$$

giving your answers in exact form.

**Question 5.** Solve the following equations, writing your answer in the form $x = a^b$:

a) $\log_3 x = 8$
b) $\log_8 x = 3$
c) $\log_5 x = 7$
d) $\log_7 x = 5$

**Question 6.** Write each of these as a single logarithm:

a) $\log_3 4 + \log_3 5$
b) $\log_5 3 + \log_5 6$
c) $\log_6 10 - \log_6 2$
d) $\log_{10} 4 - \log_{10} 8$

**Question 7.** Write each of these as a single logarithm:

a) $\log_{10} 4 + \log_{10} 3 - \log_{10} 2$
b) $\log_2 20 - \log_2 6 + \log_2 3$
c) $\log_5 9 - \log_5 10 - \log_5 3$
d) $\log_9 8 - \log_9 10 + \log_9 4$

**Question 8.** Solve the equation

$$\log_3 x - \log_3 5 = \log_3 8$$

**Question 9.** Solve the equation

$$\log_7 9 - \log_7 x = \log_7 18$$

**Question 10.** Evaluate each of these:

a) $\log_3 3$
b) $\log_8 8$
c) $2\log_5 5$
d) $-4\log_{10} 10$

**Question 11.** Write each of these in the form $\log_a b$:

a) $2\log_3 5$
b) $3\log_4 3$
c) $4\log_9 2$
d) $5\log_7 10$

**Question 12.** Write each of these in the form $\log_a b$, where $b$ is a rational number:

a) $-\log_2 3$
b) $-\log_3 2$
c) $-2\log_5 2$
d) $-3\log_8 3$

**Question 13.** Write each of these as a single logarithm:

a) $\log_b 2 + 2\log_b 5$

b) $\log_b 5 + 3\log_b 3$

c) $\log_b 10 - 4\log_b 2$

d) $\log_b 4 - 2\log_b 10$

**Question 14.**

a) Express $3\log_2 x + \log_2 k$ as a single logarithm.

b) Given that $3\log_2 x + \log_2 k = 3$, express $x$ in terms of $k$

**Question 15.**

a) Express $5\log_3 x - \log_3 m$ as a single logarithm.

b) Given that $5\log_3 x - \log_3 m = 2$, express $x$ in terms of $m$

**Question 16.** Write each of these as a single logarithm:

a) $\log_5 30 - 2\log_5 2$

b) $\log_2 9 - 3\log_2 3$

c) $\log_9 100 + 2\log_9 5$

d) $\log_3 20 + 3\log_3 10$

**Question 17.** Given that $\log_5 y^5 - \log_5 x^3 = 10$, show that $y = mx^n$, where $m$ is an integer and $n$ is a rational number.

**Question 18.** Write each of these as a single logarithm:

a) $2\log_3 4 + 4\log_3 2$
b) $3\log_5 4 - 2\log_5 2$
c) $3\log_8 5 + 2\log_8 10$
d) $4\log_4 3 - 3\log_4 9$

**Question 19.** Write each of these as a single logarithm:

a) $\frac{1}{2}\log_7 4 + \frac{1}{3}\log_7 8$

b) $\frac{1}{2}\log_{10} 81 - \frac{1}{3}\log_{10} 125$

c) $\frac{2}{3}\log_2 27 + \frac{3}{2}\log_2 9$

d) $\frac{4}{5}\log_9 32 - \frac{5}{4}\log_9 16$

**Question 20.** Write each of these as a single logarithm:

a) $4\log_3 2 + 1$

b) $3\log_2 5 - 1$

c) $2\log_5 6 + 2$

d) $3 - 2\log_4 10$

**Question 21.** Write each of these as a single logarithm:

a) $2\log_b 3 + 2$

b) $4\log_b 2 - 1$

c) $3\log_b 4 + 1$

d) $1 - 2\log_b 20$

**Question 22.** It is given that

$$\log_b p - 2\log_b q = \log_b(p - 2q)$$

Show that

$$p = \frac{2q^3}{q^2 - 1}$$

where $p > 2q > 0$

**Question 23.** It is given that

$$3\log_b q - \frac{1}{2}\log_b p = \log_b(3q - p)$$

Show that

$$p = \frac{q^6}{9q^2 - 6pq + p^2}$$

where $3q > p > 0$

**Question 24.** Find the value of:

a) $\log_b b^2 + \log_b b^5$
b) $\log_b b^4 - \log_b \frac{1}{b}$
c) $\log_b \frac{1}{b^2} + \log_b b^3$
d) $\log_b \frac{1}{b} - \log_b \frac{1}{b^3}$

**Question 25.** Show that, for $x > 0$,

$$\log_3 \frac{x^3}{9} + \log_3 27x - \log_3 x^5 = \log_3 \frac{3}{x}$$

**Question 26.** Given that $m = \log_4 x$, where $x$ is positive, write each of these in their simplest form in terms of $m$:

a) $\log_4 \sqrt{x}$

b) $\log_4 \frac{x}{4}$

c) $\log_4 \frac{16}{x^2}$

**Question 27.** It is given that

$$\log_b y = 3\log_b 2 + \log_b 5 - \frac{3}{2}$$

Write $y$ in terms of $b$ without any logarithms.

**Question 28.** It is given that

$$\log_k y = \log_k 8 - 2\log_k \frac{1}{2} + \frac{3}{4}$$

Write $y$ in terms of $k$ without any logarithms.

**Question 29.** Write each of these as a single logarithm:

a) $2\log_b p + 3\log_b q + 2\log_b r$
b) $3\log_b p + 5\log_b q - 3\log_b r$
c) $5\log_b p - 2\log_b q + 4\log_b r$
d) $8\log_b p - 6\log_b q - 9\log_b r$

**Question 30.** Write each of these as a single logarithm:

a) $\log_7 30 - (\log_7 2 + \log_7 3)$
b) $\log_8 15 - 2(\log_8 6 - \log_8 2)$
c) $\log_6 60 - 3(\log_6 5 + \log_6 2)$
d) $\log_{10} 50 - 2(\log_{10} 80 - \log_{10} 16)$

**Question 31.** Write each of these as a single logarithm:

a) $\log_b p - (\log_b q + \log_b r)$
b) $\log_b p - 2(\log_b q - \log_b r)$
c) $\log_b p - 3(\log_b q + \log_b r)$
d) $2\log_b p - 2(\log_b q - 2\log_b r)$

**Question 32.** It is given that $x = \log_b 3$, $y = \log_b 5$ and $z = \log_b 7$. Write each of these in terms of $x, y$ and $z$:

a) $\log_b 15$
b) $\log_b 9$
c) $\log_b 21$
d) $\log_b \frac{49}{25}$

**Question 33.** By writing $\log_3 4 \times \log_3 64$ in terms of $\log_3 4$, show that

$$\log_3 4 \times \log_3 64 > 3$$

**Question 34.** Showing your working, write each of these in a form not involving logarithms:

a) $\log_{100} 10 + \log_{10} 100$
b) $\log_2 8 - \log_8 2$
c) $\log_9 81 + \log_3 \frac{1}{3}$
d) $\log_{25} \frac{1}{625} + \log_5 625$

**Question 35.** A right-angled triangle ABC has hypotenuse AC. The length of AB is $(\log_5 16)$ cm and the length of BC is $(\log_5 64)$ cm. Without the use of a calculator, show that the area of the triangle is larger than 0.75 $\text{cm}^2$.

**Question 36.** Using $y = 2^x$ as a substitution, show that

$$4^x - 7 \times 2^{x+1} + 45 = 0$$

can be written as

$$y^2 - 14y + 45 = 0$$

Hence show that $4^x - 7 \times 2^{x+1} + 45 = 0$ has two solutions of the form $\log_2 k$.

**Question 37.** By writing $\log_4 3 \times \log_4 9 \times \log_4 81$ in terms of $\log_4 3$, show that

$$\log_4 3 \times \log_4 9 \times \log_4 81 < 8$$

**Question 38.** Using $y = 2^{2x}$ as a substitution, show that

$$16^x + 2^{2x+1} - 99 = 0$$

can be written as

$$y^2 + 2y - 99 = 0$$

Hence show that $16^x + 2^{2x+1} - 99 = 0$ has only one solution of the form $\log_2 k$.

**Question 39.** Solve the following equations, writing your answers in logarithmic form:

a) $2^{2x} - 8 \times 2^x + 15 = 0$

b) $3^{2x} - 9 \times 3^x + 20 = 0$

c) $4^{2x} - 6 \times 4^x - 7 = 0$

d) $3^{2x} - 6 \times 3^x - 40 = 0$

**Question 40.** Solve the following equations, writing your answers in logarithmic form:

a) $2^{2x} - 5 \times 2^{x+1} + 21 = 0$

b) $5^{2x} - 3 \times 5^{x+1} + 50 = 0$

c) $4^{2x} + 2^{2x+1} - 15 = 0$

d) $4^{2x} - 5 \times 2^{2x+1} + 2^{2x+3} - 80 = 0$

**Question 41.** Solve the equation

$$8^x - 2^{2x+2} + 3 \times 2^x = 0$$

**Question 42.** Solve the following equations, writing your answers in logarithmic form:

a) $3^{2x+1} - 13 \times 3^x - 10 = 0$

b) $16^{x+1} - 139 \times 4^x - 45 = 0$

c) $4^{2x+3} - 9 \times 2^{2x+5} = 0$

d) $2^{4x+7} + 2^{2x+3} - 3 \times 4^{x+2} - 3 = 0$

**Question 43.** Show that the solution to the equation

$$5^{x-3} = 2^{x+4}$$

can be written in the form

$$\frac{4\log_5 2 + 3}{1 - \log_5 2}$$

**Question 44.** Show that the solution to the equation

$$8^{x+2} = 3^{2-x}$$

can be written in the form

$$\frac{2 - 2\log_3 8}{\log_3 8 + 1}$$

**Question 45.** Show that the solution to the equation

$$3^{x+1} = 8^{x-1}$$

can be written in the form

$$\frac{\log_{10} 8 + \log_{10} 3}{\log_{10} 8 - \log_{10} 3}$$

**Question 46.** Solve each of these equations by taking logarithms of base 3 of both sides:

a) $3^x = 4^{x-1}$
b) $3^{x+5} = 5^{2-x}$
c) $8^{x-3} = 3^{x+8}$
d) $10^{5-x} = 3^{x-6}$

**Question 47.** Solve each of these equations by taking logarithms of base 2 of both sides:

a) $5^x = 9^{x+2}$

b) $3^{x-2} = 7^{3-x}$

c) $9^{x+4} = 10^{4-x}$

d) $5^{8-x} = 6^{4-x}$

**Question 48.**

a) Given that

$$2\log_{10}(2-x) = \log_{10}(28-9x)$$

show that $x^2 + 5x - 24 = 0$

b) Hence show that

$$2\log_{10}(2-x) = \log_{10}(28-9x)$$

has only one solution.

**Question 49.** Solve the equation

$$\log_5(8x-1) - \log_5(2-x) = 2$$

**Question 50.** Julia is trying to solve the equation

$$3(\log_8 x)^2 + 11\log_8 x - 20 = 0$$

Julia's attempt is shown below:

$$\begin{aligned} 3(\log_8 x)^2 + 11\log_8 x - 20 &= 0 && \text{Line 1}\\ 6\log_8 x + 11\log_8 x - 20 &= 0 && \text{Line 2}\\ 17\log_8 x - 20 &= 0 && \text{Line 3}\\ \log_8 x &= \tfrac{20}{17} && \text{Line 4}\\ x &= (\tfrac{20}{17})^8 && \text{Line 5} \end{aligned}$$

a) Identify the two errors made by Julia.

b) Solve the equation

$$3(\log_8 x)^2 + 11\log_8 x - 20 = 0$$

giving your answers in exact form.

**Question 51.** George is trying to solve the equation

$$\log_3 \frac{1}{x} + \frac{1}{4\log_3 x} = 0$$

George's attempt is shown below:

$$\log_3 \tfrac{1}{x} + \tfrac{1}{4\log_3 x} = 0 \quad \text{Line 1}$$
$$\log_3 \tfrac{1}{x} + \tfrac{1}{4}\log_3 \tfrac{1}{x} = 0 \quad \text{Line 2}$$
$$\tfrac{5}{4}\log_3 \tfrac{1}{x} = 0 \quad \text{Line 3}$$
$$-\tfrac{5}{4}\log_3 x = 0 \quad \text{Line 4}$$
$$x = 3 \quad \text{Line 5}$$

a) Identify the two errors made by George.

b) Solve the equation

$$\log_3 \frac{1}{x} + \frac{1}{4\log_3 x} = 0$$

giving your answers in exact form.

**Question 52.**

a) Show that the equation

$$2\log_3 x = \log_3(mx - 4) + 4$$

where $m$ is a positive constant, can be expressed in the form

$$x^2 - 81mx + 324 = 0$$

b) Given that $x^2 - 81mx + 324 = 0$ has only one root, find the value of $m$

**Question 53.** Solve $2^{2^x} = 5$, writing your answer in an exact form.

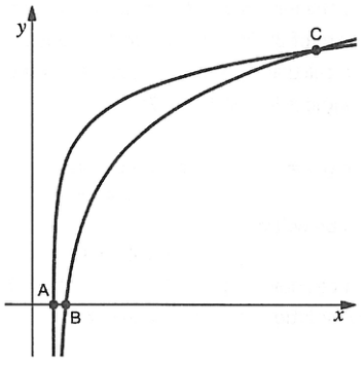
**Question 54.** The graph below shows two curves that intersect at the point C. The curve that passes through A has equation

$$y = \log_5(6x - 15) + 1$$

and the curve that passes through B has equation

$$y = 2\log_5(x - 3)$$

Determine the exact coordinates of the points A, B and C.



**Question 55.** The graph below shows two curves that intersect at the points A and B. The curve that passes through C has equation

$$y = 2\log_4(x+6)$$

and the other curve has equation

$$y = \log_4(x+48) - \log_4(2-x)$$

Determine the exact coordinates of the points A, B and C.

